Crime Record Management System
for Bengaluru

Project Draft — Database Report

DBMS Project	Urban Law Enforcement	Cyber + Traditional Crime

1. Project Description

The Crime Record Management System (CRMS) is a relational database-driven application designed to support efficient management of crime records in an urban setting like Bengaluru. The system addresses the dual challenge of handling both conventional crimes — theft, assault, and vandalism — and emerging digital offences such as cyber fraud, phishing, and online scams, which have seen a sharp rise in India's fastest-growing metro.

Built on a normalised schema with three core tables — cases, officers, and the junction table case_officer — the system captures the full lifecycle of a case from initial complaint registration to final resolution. Each case record holds structured attributes including crime type, geographic location, complaint mode (online or offline), and a tracked status field (Active / Solved / Closed), enabling both routine queries and analytical reporting.

The project demonstrates core DBMS concepts in a practical, real-world context: entity relationships, normalisation, referential integrity via foreign keys, enum constraints for controlled data entry, and multi-table JOIN operations to retrieve comprehensive case views.

2. Feature List

The following eight features define the functional scope of the system:

#	Feature	Description
1	**Case Registration**	Log new crime cases with title, description, crime type, location (area in Bengaluru), complaint mode (online / offline), and auto-timestamped date of filing.
2	**Case Lifecycle Tracking**	Track every case through its lifecycle with a three-state status field: Active, Solved, or Closed. Status can be updated without deleting the record, preserving full history.

#	Feature	Description
3	**Officer Management**	Maintain a dedicated roster of police officers with officer ID, name, and rank/designation. Officers are stored independently to eliminate redundancy across case records.
4	**Case-Officer Assignment**	Assign multiple officers to a single case, or the same officer to multiple cases, via a junction table — correctly modelling the many-to-many relationship in real police workflows.
5	**Filtered Case Retrieval**	Query and filter cases by status, crime type, or location. Supports targeted lookups such as all Active Cyber Fraud cases in Whitefield for operational decision-making.
6	**Crime Type Categorisation**	Classify cases under crime types including Cyber Fraud, Theft, Assault, and others — enabling statistical aggregation and category-based reporting across Bengaluru districts.
7	**Data Integrity Enforcement**	Enum constraints on status and complaint_mode, NOT NULL on critical fields, and foreign key relationships ensure only valid, consistent data enters the system.
8	**JOIN-Based Case Reports**	Generate combined views linking case details with assigned officers using multi-table JOINs — producing actionable reports without data duplication in the underlying schema.

3. Schema at a Glance

The system uses three normalised tables connected through primary and foreign key relationships:

`cases`	`case_officer`	`officers`
PK: case_id status, crime_type, location, complaint_mode, date_reported	Junction Table (M:M) FK: case_id FK: officer_id	PK: officer_id name, rank

Enum constraints: status (Active / Solved / Closed) | complaint_mode (Online / Offline)

4. Supported Database Operations

Operation	Description
`INSERT`	Add new cases, register officers, and create officer-case assignments.
`SELECT`	Retrieve all cases, filter by status, crime type, or location, and view case-officer mappings via JOIN.

Operation	Description
`UPDATE`	Modify case status, update descriptive fields, and reassign officers as investigation progresses.
`DELETE`	Optionally remove records. Recommended approach is status change to Closed to retain full historical data.
`JOIN`	Combine cases and officers into unified views to support reporting and workload management across the station.

5. User Definition

The system serves four distinct user types, each with different access needs and interaction patterns:

5.1 Investigating Officer

IO **Investigating Officer**	A police officer directly responsible for handling one or more active cases. Needs to log new complaints, update case progress, and view the list of cases assigned to them. May not have deep technical knowledge — expects straightforward forms and filtered views. **Key needs:** • Register new cases (online and offline) • Update case status as investigation progresses • View all cases currently assigned to them • Add or update case descriptions and details

5.2 Station Inspector / Supervisor

SI **Station Inspector**	A senior officer overseeing multiple investigators and cases at a police station. Requires a high-level view of all active cases, the ability to assign officers, and summary reports by crime type and area. Uses the system for resource allocation and workload balancing. **Key needs:** • Assign officers to cases via case_officer table • Filter and view cases by location, crime type, and status • Monitor overall case lifecycle across the station • Generate JOIN-based reports for operational decisions

5.3 Data / Records Administrator

DA **Data Administrator**	A non-operational staff member responsible for maintaining data integrity, managing officer records, and ensuring the database remains clean and consistent. Handles bulk updates, flags mismatches in enum values, and maintains the officer roster. **Key needs:** • Manage and update the officers table • Enforce enum constraints and NOT NULL conditions • Perform bulk status updates on closed cases • Maintain schema integrity and referential consistency

5.4 Citizen / Complainant (Indirect User)

CA **Citizen / Complainant**	A member of the public who files a complaint, either in person at the station (offline) or via an online portal. They are not direct database users but are the source of new case records. The complaint_mode field distinguishes how their complaint entered the system. **Key needs:** • File complaints via online portal (complaint_mode = Online) • File complaints in-person at station (complaint_mode = Offline) • Indirect source of all case records entered into the system

Note on Pending Areas

The highlighted areas in the original PDF — schema design, complaint_mode handling, and officer assignment logic — are marked for review and may be revised once the full implementation scope is confirmed.